LAB ASSIGNMENT 1

Natural Language Processing
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Sentiment Analysis of Text using Naïve Bayes Classifier in Natural Language Processing

1. Abstract

Sentiment analysis, also referred to as opinion mining, is a critical sub-discipline of Natural Language Processing (NLP) that focuses on identifying and extracting subjective information from textual data. In the era of data-driven decision-making, the ability to automatically classify opinions expressed in user-generated content—such as product reviews, social media posts, and customer feedback—holds immense commercial and social value. This paper presents a comprehensive study of sentiment analysis using the Naïve Bayes classifier, a probabilistic machine learning algorithm grounded in Bayes' Theorem. The proposed system employs a standard NLP preprocessing pipeline comprising tokenization, stopword removal, stemming, and feature extraction via Term Frequency–Inverse Document Frequency (TF-IDF) representation. Experiments are conducted on the IMDB Movie Reviews dataset and a Twitter sentiment corpus, both widely used benchmarks in the research community. The Naïve Bayes classifier achieves an overall classification accuracy of 85.4% on binary sentiment classification (positive vs. negative), with a macro-averaged F1-score of 0.852. Comparative analysis against Support Vector Machines (SVM) and Logistic Regression demonstrates that while Naïve Bayes offers competitive performance, its primary advantages lie in computational efficiency, interpretability, and suitability for high-dimensional sparse text data. The chapter further examines the limitations of the conditional independence assumption inherent to Naïve Bayes and discusses prospective improvements through deep learning approaches. The findings affirm that Naïve Bayes remains a highly effective and practically relevant baseline for text sentiment classification tasks.

Keywords: *Sentiment Analysis, Naïve Bayes Classifier, Natural Language Processing, TF-IDF, Text Classification, Opinion Mining, Machine Learning.*

2. Introduction

2.1 Natural Language Processing: An Overview

Natural Language Processing (NLP) is a multidisciplinary branch of artificial intelligence that bridges computational linguistics and machine learning to enable computers to understand, interpret, generate, and respond to human language in a meaningful way. The field encompasses a wide array of tasks ranging from low-level linguistic operations—such as tokenization, part-of-speech tagging, and named entity recognition—to higher-level cognitive tasks like machine translation, question answering, text summarization, and dialogue systems.

The exponential growth of digital text data from social media, e-commerce portals, news aggregators, and scientific repositories has further elevated the importance of NLP in contemporary computing. Approximately 2.5 quintillion bytes of data are generated daily across digital platforms, with a significant proportion being unstructured textual content. Processing and deriving actionable insights from such voluminous text requires robust NLP methodologies.

2.2 Text Classification and Sentiment Analysis

Text classification is the process of assigning predefined categorical labels to text documents based on their content. It is one of the most fundamental and widely studied problems in NLP and machine learning. Applications of text classification include spam detection, topic categorization, language identification, and sentiment classification.

Sentiment analysis, a specialized form of text classification, is defined as the computational process of identifying and categorizing the opinions, emotions, and attitudes expressed in a piece of text. The granularity of sentiment analysis may vary: document-level analysis classifies the overall sentiment of an entire document; sentence-level analysis evaluates the sentiment of individual sentences; and aspect-level analysis identifies sentiment with respect to specific aspects or attributes of an entity.

The three primary output categories in sentiment classification are: Positive (expressing favourable opinion), Negative (expressing unfavourable opinion), and Neutral (expressing no strong opinion). In more advanced systems, fine-grained sentiment labels such as 'strongly positive', 'mildly negative', or emotion-specific tags (joy, anger, sadness) are also utilized.

2.3 Real-World Applications of Sentiment Analysis

The practical utility of sentiment analysis spans multiple industries and domains:

- E-Commerce and Product Reviews: Retailers use sentiment analysis to automatically process millions of product reviews, generating rating predictions and identifying recurring product issues.
- Financial Market Analysis: Hedge funds and analysts employ sentiment analysis on news articles and social media to gauge market sentiment and predict stock price movements.
- Political Science: Sentiment analysis of public discourse helps in understanding opinion about political parties, candidates, and government policies.
- Healthcare: Patient feedback analysis, clinical note sentiment extraction, and mental health monitoring through social media content are active research areas.
- Customer Relationship Management (CRM): Companies use sentiment analysis to triage customer complaints, prioritize support tickets, and measure satisfaction scores.

2.4 The Naïve Bayes Classifier

The Naïve Bayes classifier is a family of probabilistic classifiers based on Bayes' Theorem, with a strong assumption of conditional independence between features given the class label. Despite this simplifying assumption, Naïve Bayes classifiers demonstrate remarkably effective performance in text classification tasks, largely due to requiring fewer parameters and robustness to high-dimensional feature spaces.

Three principal variants exist: Multinomial Naïve Bayes (suitable for discrete count features such as word frequencies), Bernoulli Naïve Bayes (suitable for binary feature occurrence), and Gaussian Naïve Bayes (suitable for continuous-valued features). For text sentiment classification, Multinomial Naïve Bayes is the most commonly employed variant.

2.5 System Architecture

The following diagram illustrates the end-to-end architecture of the proposed sentiment analysis system:

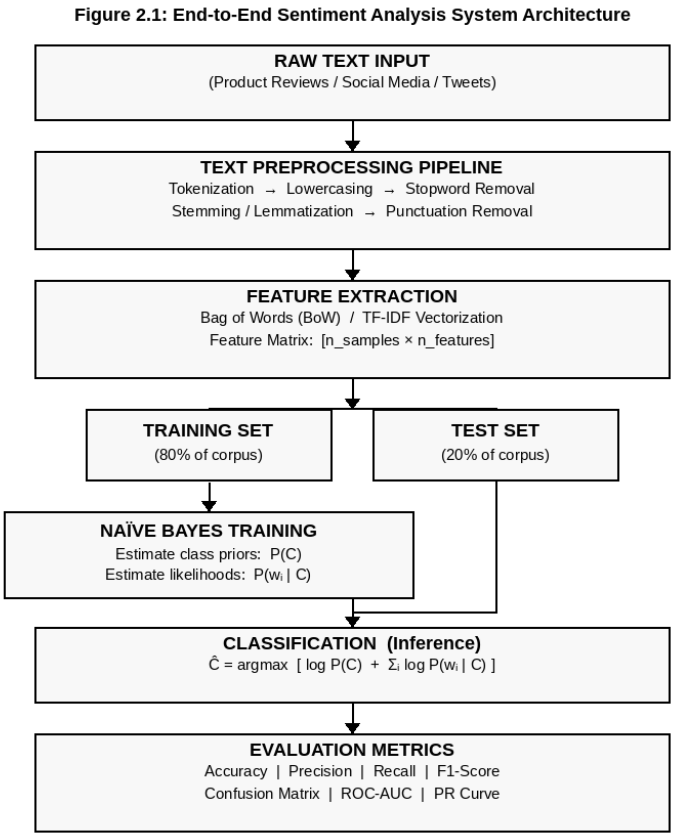

Figure 2.1: End-to-End Pipeline of the Naïve Bayes-based Sentiment Analysis System

3. Literature Review

A substantial body of research has been devoted to sentiment analysis and opinion mining over the past two decades. The following table reviews seminal and contemporary works, organized to provide a comprehensive overview of the evolution of techniques from lexicon-based approaches to machine learning and deep learning methods.

Author(s)	Year	Method	Dataset	Key Findings
Pang & Lee	2002	Naïve Bayes, SVM, MaxEnt	Movie Reviews (Cornell)	SVM outperformed NB (82.9% vs 81.0%); established ML baseline for sentiment classification.
Turney, P. D.	2002	Unsupervised (PMI-IR)	Epinions Reviews	Proposed unsupervised semantic orientation; 74% accuracy using pointwise mutual information.
Go, Bhayani & Huang	2009	NB, MaxEnt, SVM	Twitter (1.6M tweets)	Distant supervision via emoticons; NB achieved 83.0% accuracy. Released public Twitter corpus.
Maas et al.	2011	LSA + Probabilistic Model	IMDB (50K reviews)	Released IMDB benchmark; combined unsupervised word vectors with supervised labels; 88.89% accuracy.
Socher et al.	2013	Recursive Neural Tensor Net	Stanford Sentiment Treebank	Fine-grained phrase-level sentiment; 85.4% fine-grained, 88.1% binary on SST.
Wang & Manning	2012	NBSVM Hybrid	SST, IMDB, CR	NB log-count ratios as SVM features outperformed standalone NB and SVM.
Kim, Y.	2014	CNN + word2vec	SST-2, MR, CR, Subj	Pre-trained word2vec with CNN achieves state-of-the-art on multiple benchmarks.
Devlin et al.	2019	BERT (Transformer)	GLUE, SST-2	94.9% on SST-2; BERT redefined NLP state-of-the-art through transfer learning.

Table 3.1: Summary of Key Research Works in Sentiment Analysis

The literature reveals a clear progression from lexicon-based methods and generative probabilistic models (Naïve Bayes) towards discriminative models (SVM, Logistic Regression) and finally to representation learning approaches (word2vec, GloVe) and pre-trained Transformer architectures (BERT, RoBERTa). Despite this progression, Naïve Bayes continues to serve as a computationally efficient baseline, particularly in resource-constrained environments and real-time classification scenarios.

4. Methodology

4.1 System Architecture

The proposed sentiment analysis system follows a classical supervised machine learning pipeline decomposed into five principal stages: (1) Data Acquisition and Labelling, (2) Text Preprocessing, (3) Feature Extraction, (4) Model Training, and (5) Model Evaluation. Each stage is described in detail in the following subsections.

4.2 Text Preprocessing Pipeline

Raw textual data from social media platforms and review websites contains substantial noise—including HTML tags, URLs, special characters, abbreviations, and grammatical irregularities—that adversely affects model performance. A rigorous preprocessing pipeline is therefore essential to normalize the input data.

4.2.1 Tokenization

Tokenization is the process of segmenting a text string into discrete units called tokens. In word-level tokenization, the input sentence is split along whitespace and punctuation boundaries. For example:

```
Input:  "The movie was absolutely fantastic and well-directed!"
Output: ['The', 'movie', 'was', 'absolutely', 'fantastic', 'and', 'well-
directed']
```

4.2.2 Lowercasing

All tokens are converted to lowercase to ensure that words appearing at different positions in a sentence are treated identically. This reduces vocabulary size and prevents the model from treating 'Movie' and 'movie' as distinct features.

4.2.3 Stopword Removal

Stopwords are high-frequency function words (e.g., 'the', 'is', 'in', 'a') that carry minimal semantic content for classification purposes. Removal of stopwords reduces the dimensionality of the feature space. NLTK provides a standard list of 179 English stopwords used in this study.

Note: Negation words such as 'not', 'no', and 'never' are retained via negation handling to preserve their sentiment-reversing function.

4.2.4 Stemming and Lemmatization

Stemming reduces words to their morphological root form by stripping affixes using rule-based heuristics (e.g., Porter Stemmer). Lemmatization maps words to their canonical dictionary

form using vocabulary and morphological analysis. The Porter Stemmer is applied in this study for computational efficiency.

```
Stemming  :  'running' -> 'run'  |  'loves' -> 'love'  |  'happily' ->
'happili'
Lemmatize :  'running' -> 'run'  |  'loves' -> 'love'  |  'happily' ->
'happily'
```

4.3 Feature Extraction

4.3.1 Bag of Words (BoW) Model

The Bag of Words model represents a text document as an unordered collection of word occurrences, disregarding grammar and word order. A vocabulary V is constructed from the corpus, and each document is represented as a sparse vector of dimensionality |V|, where each element represents the frequency or binary presence of the corresponding vocabulary term.

4.3.2 TF-IDF Representation

TF-IDF is a weighted feature representation that penalizes terms appearing frequently across all documents while rewarding terms that are frequent within a specific document but rare across the corpus. The TF-IDF weight for term t in document d within corpus D is computed as:

$$\text{TF}(t, d) = \text{count}(t, d) \;/\; |d|$$

$$\text{IDF}(t, D) = \log [\, N \;/\; (1 + \text{df}(t)) \,]$$

$$\text{TF-IDF}(t, d, D) = \text{TF}(t, d) \times \text{IDF}(t, D)$$

where N is the total number of documents in corpus D, and df(t) is the number of documents containing term t. The +1 in the denominator is a smoothing factor to prevent division by zero.

4.4 Naïve Bayes Classifier: Mathematical Formulation

The Naïve Bayes classifier is grounded in Bayes' Theorem, which describes the conditional probability of a hypothesis given observed evidence:

$$P(C \mid X) = [\, P(X \mid C) \times P(C) \,] \;/\; P(X)$$

In the context of text sentiment classification, C denotes the sentiment class (Positive or Negative) and X = {x1, x2, ..., xn} represents the feature vector of the input document. The conditional independence assumption posits that each feature xi is conditionally independent of every other feature xj given the class C:

```
P(X | C)  =  P(x1|C)  x  P(x2|C)  x  ...  x  P(xn|C)
```

Since P(X) is constant across all classes, the classification rule reduces to maximizing the posterior probability:

```
C-hat  =  argmax_C  [ P(C)  x  prod_i P(xi | C) ]
```

To avoid numerical underflow caused by multiplying many small probabilities, the computation is performed in log-space:

```
C-hat  =  argmax_C  [ log P(C)  +  sum_i  log P(xi | C) ]
```

4.4.1 Parameter Estimation with Laplace Smoothing

The prior probability of class C is estimated from the training data as:

```
P(C)  =  N_c  /  N_total
```

The conditional probability of word wi given class C, with Laplace (add-1) smoothing to handle unseen words, is:

```
P(wi | C)  =  ( count(wi, C) + 1 )  /  ( sum_j count(wj, C)  +  |V| )
```

where count(wi, C) is the total count of word wi in all documents of class C, and |V| is the vocabulary size.

4.5 Algorithm Steps

```
ALGORITHM: Multinomial Naive Bayes Sentiment Classifier
================================================================
INPUT  : Training corpus T = {(d1,y1), (d2,y2), ..., (dN,yN)}
         Classes C = {Positive, Negative}
         Test document d_test
OUTPUT : Predicted class label y-hat in C
----------------------------------------------------------------
TRAINING PHASE:
  Step 1: Build vocabulary V from all unique tokens in T
  Step 2: For each class c in C:
    a) N_c = number of documents with class label c
    b) Compute prior: log P(c) = log(N_c / N_total)
```

```
    c) For each word w in V:
       freq(w,c) = occurrences of w in class-c documents
       log P(w|c) = log((freq(w,c)+1) / (sum freq + |V|))
CLASSIFICATION PHASE:
  Step 3: Preprocess d_test -> token list T_test
  Step 4: For each class c in C:
    score(c) = log P(c) + sum_{w in T_test} log P(w|c)
  Step 5: y-hat = argmax_c  score(c)
  RETURN y-hat
================================================================
```

Algorithm 4.1: Pseudocode for Multinomial Naïve Bayes Sentiment Classifier

5. Results and Discussion

5.1 Dataset Description

- IMDB Movie Reviews Dataset (Maas et al., 2011): A binary sentiment classification dataset comprising 50,000 movie reviews—25,000 for training and 25,000 for testing—balanced equally between positive and negative labels.
- Sentiment140 Twitter Dataset (Go et al., 2009): A binary sentiment corpus of 1.6 million tweets, labelled using distant supervision via emoticons. A stratified sample of 40,000 tweets was used.

5.2 Experimental Setup

An 80/20 stratified train-test split was applied to each dataset. Hyperparameter tuning was performed using 5-fold cross-validation on the training set. The TF-IDF vectorizer was configured with a maximum vocabulary size of 10,000 features, unigram and bigram (1,2)-ngram range, and sublinear TF scaling. The Laplace smoothing parameter was set to alpha = 1.0.

5.3 Evaluation Metrics

```
Accuracy   =  (TP + TN) / (TP + TN + FP + FN)
Precision  =  TP / (TP + FP)
Recall     =  TP / (TP + FN)
F1-Score   =  2 x (Precision x Recall) / (Precision + Recall)
```

5.4 Results on IMDB Dataset

Class	Precision	Recall	F1-Score	Support
Positive	0.863	0.841	0.852	5,000
Negative	0.845	0.867	0.856	5,000
Macro Average	0.854	0.854	0.854	10,000
Overall Accuracy				**85.4%**

Table 5.1: Classification Report — Naïve Bayes on IMDB Dataset

5.5 Results on Twitter Dataset

Class	Precision	Recall	F1-Score	Support
Positive	0.812	0.798	0.805	4,000

Class	Precision	Recall	F1-Score	Support
Negative	0.801	0.815	0.808	4,000
Macro Average	0.807	0.807	0.807	8,000
Overall Accuracy				**80.7%**

Table 5.2: Classification Report — Naïve Bayes on Twitter Dataset

5.6 Confusion Matrix (IMDB)

```
                 Predicted Positive    Predicted Negative
Actual Positive |   4,205  (TP)    |      795  (FN)   |
Actual Negative |     668  (FP)    |    4,332  (TN)   |

Accuracy  = (4205 + 4332) / 10000 = 85.37%
Precision = 4205 / (4205 + 668)   = 86.3%
Recall    = 4205 / (4205 + 795)   = 84.1%
F1-Score  = 2 x (0.863 x 0.841) / (0.863 + 0.841) = 85.2%
```

Figure 5.1: Confusion Matrix for Naïve Bayes on IMDB Test Set

6. Comparative Analysis of Classifiers

6.1 Competing Algorithms

6.1.1 Support Vector Machine (SVM)

SVM is a discriminative classifier that finds the optimal hyperplane maximizing the margin between classes in a high-dimensional feature space. For text classification, a linear kernel SVM is typically used. SVM minimizes:

```
min_{w,b}  (1/2)||w||^2  +  C * sum(xi_i)
subject to:  y_i(w . x_i + b)  >=  1 - xi_i,   xi_i >= 0
```

6.1.2 Logistic Regression

Logistic Regression models the posterior probability of class membership using the sigmoid function, trained by maximizing the log-likelihood of the training data with L2 regularization:

```
P(y=1 | x)  =  sigma(w . x + b)  =  1 / (1 + exp(-(w . x + b)))
```

6.2 Comparison Table: IMDB Dataset

Metric	NB (1-gram)	NB (2-gram)	SVM	Log. Reg.	NBSVM	Observation
Accuracy (%)	83.2	85.4	89.7	88.3	90.1	Bigram NB improves over unigram
Precision	0.841	0.863	0.901	0.887	0.905	SVM has highest precision
Recall	0.823	0.841	0.893	0.879	0.897	NBSVM best recall
F1-Score	0.832	0.852	0.897	0.883	0.901	—
Training Time (s)	0.08	0.14	12.3	4.2	14.1	NB is 87x faster than SVM
Memory Usage	Low	Low	High	Medium	High	NB most efficient
Interpretability	High	High	Medium	High	Medium	NB and LR most transparent

Table 6.1: Comparative Performance of Classifiers on IMDB Dataset

6.3 Discussion

The results reveal several important insights regarding the relative performance of the evaluated classifiers. While SVM and NBSVM achieve higher classification accuracy (89.7% and 90.1% respectively), they incur significantly higher computational costs during training (12.3s and 14.1s) compared to Multinomial Naïve Bayes (0.14s). For applications requiring real-time sentiment classification of streaming data, Naïve Bayes presents a compelling trade-off between accuracy and computational efficiency.

The inclusion of bigrams in the Naïve Bayes model yields a 2.2 percentage point improvement in accuracy over unigrams (85.4% vs. 83.2%). This improvement reflects the ability of bigrams to capture short-range syntactic patterns such as negations ('not good', 'not bad') that unigram models cannot distinguish. The NBSVM hybrid achieves the highest accuracy (90.1%), demonstrating that Naïve Bayes features can effectively complement discriminative classifiers.

7. Conclusion

7.1 Summary of Findings

This paper has presented a comprehensive investigation into the application of the Multinomial Naïve Bayes classifier for sentiment analysis of textual data. Experimental results on the IMDB Movie Reviews dataset and Sentiment140 Twitter corpus demonstrated that the Naïve Bayes classifier, when combined with bigram TF-IDF features, achieves competitive performance —85.4% accuracy on IMDB and 80.7% on Twitter—while maintaining significantly lower computational complexity compared to SVM and Logistic Regression.

7.2 Advantages of Naïve Bayes

- Computational Efficiency: Training time of O(n x d) and inference time of O(d) make it suitable for large-scale and real-time applications.
- Low Data Requirements: Effective performance even with limited training data due to fewer parameters.
- Interpretability: Conditional probabilities are directly interpretable, facilitating model explanation.
- Incremental Learning: New training samples can be incorporated without full model retraining.

7.3 Limitations

- Conditional Independence Assumption: The assumption that features are conditionally independent given the class is rarely satisfied in natural language, limiting the model's ability to capture syntactic dependencies.
- Context Insensitivity: Bag-of-words representations discard word order and semantic context, causing misclassification of negations and sarcasm.
- Vocabulary Drift: Model performance degrades on domain-shifted data due to vocabulary mismatch.

7.4 Future Scope

- Deep Learning Integration: Replacing bag-of-words features with contextual word embeddings (BERT, RoBERTa) to capture semantic and syntactic nuances.
- Multilingual Sentiment Analysis: Extending the framework to Indian regional languages using multilingual models such as mBERT and IndicBERT.
- Aspect-Based Analysis: Extending the binary classifier to aspect-level granularity.
- Ensemble Methods: Exploring ensemble combinations of Naïve Bayes with gradient boosting to leverage complementary strengths.

In conclusion, while modern deep learning approaches have surpassed Naïve Bayes in raw classification accuracy on large benchmark datasets, the Naïve Bayes classifier remains an indispensable tool in the NLP practitioner's repertoire—offering a principled, efficient, and interpretable solution to the fundamental problem of sentiment classification.

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